

EDUCATION

U Institute of Engineering and Technology
Bachelor of Engineering in Computer Science & Technology

CPI - 9.1/10

EXPERIENCE

Tuvoc technologies private limited
AI/ML Engineer

Oct 2024 - Present

- Built an **end-to-end RAG-based multimodal chatbot** (text + image) using **LLMs**, supporting **freemium access**, trained over **1000+** documents, achieving high response accuracy and strong contextual retrieval performance.
IBM Professional Data Science Certification holder.
- Selected through **client interview** and worked **on-site in Bangalore** to develop a **LangGraph-based agentic chatbot** for **automatic ticket creation**, integrating a **custom-trained BERT classification model**, serving **~10K requests/day** with **85%+** classification accuracy.
- Currently designing and implementing an **agent-based issue resolution system**, where specialized agents autonomously **analyze, route, and resolve classified tickets**, reducing manual intervention and improving resolution efficiency.
- Implemented Cooperative SQL Generation (**CSMA-SQL**) using **multi-functional LLM agents**, enabling accurate SQL generation over **large, segmented enterprise databases** whose schemas exceed single-context LLM limits.

KEY PROJECTS

Agentic Ticket Classification & Resolution System | Python, LangGraph, BERT, Aws bedrock model, Langfuse

- Developed a **LangGraph-based agentic chatbot** for **automatic ticket classification and creation**, integrating a **custom-trained BERT model**.
- Achieved **85%+ classification accuracy** while serving **~10K requests/day** in a production-like environment.
- Currently designing an **agent-based issue resolution system**, where specialized agents autonomously **analyze, route, and resolve classified tickets**, significantly reducing manual intervention and improving resolution efficiency.

Large Language Model from Scratch | PyTorch, Transformers, Deep Learning , Hugging Face

- Implemented a **GPT-style Transformer language model** from scratch in **PyTorch** with **~124M parameters**.
- Built core components including **self-attention, embeddings, causal masking, and feed-forward layers**, with extensive experimentation and hyperparameter tuning.
- Exported model to **ONNX (190 MB)**, enabling **~33 ms CPU inference latency** and **~12 ms GPU latency** per sentence

Multimodal RAG-Based AI Chatbot Platform | RAG, Vision Models, Qdrant, SQL, Stripe, Fastapi, Langchain, Langsmith, Grafana

- Built an **end-to-end RAG-based multimodal chatbot** supporting **text and image inputs**, using **LLMs** for reasoning and response generation.
- Indexed and retrieved knowledge from **1000+ documents**, enabling high-accuracy answers with strong contextual grounding.
- Designed a **freemium access model**, balancing performance, cost, and scalability.
- Achieved high response **accuracy** and **robust** retrieval quality through optimized chunking, embeddings, and prompt strategies.

SKILLS

- Languages:** Python, Java, C++, JavaScript
- Database & Vector Db :** Milvus, Qdrant, ChromaDB, PostgreSQL, MySQL, MongoDB
- Frameworks & Library :** LangChain, LangGraph, LlamaIndex, CrewAI , Hugging Face , PyTorch, ONNX
- MLOps & Monitoring:** vLLM Inference, Runpod, RAGAs, Langfuse, Ray (Basics) , DVC, MLflow
- Tools & Technologies:** Git, GitHub, VS Code, Postman, Linux, Docker, GitHub Actions
- Other Skills:** Problem Solving Skills, Verbal Communication, Team Leadership, Attention to Detail

CERTIFICATIONS

- IBM Data Science Professional Certificate** | [Link](#)
Comprehensive training in **Python, SQL, data analysis, machine learning, visualization, and generative AI**, including hands-on projects and a capstone focused on real-world data science workflows.
- Mathematics for Machine Learning & Data Science Specialization by DeepLearning.AI** | [Link](#)
Strong foundation in **linear algebra, calculus, probability, and statistics**, with emphasis on mathematical concepts essential for machine learning and deep learning models.
- IBM Back-End Development Professional Certificate** | [Link](#)
Covers **Python, Django, Flask, SQL, Linux, Git, Docker, microservices, serverless architectures, application security, and observability**, culminating in a full-stack back-end capstone project.